Complete Study Notes: Computer Hardware & Digital Security

PART 1: COMPUTER HARDWARE & SYSTEM COMPONENTS

1. The System Unit

- **Definition:** The System Unit is a case that contains the electronic components of the computer used to process data.
- **Internal Components of a Desktop PC System Unit:**
    - Drive bay(s)
    - Power supply
    - Sound card
    - Video card
    - Processor
    - Memory
- **The Motherboard:**
    - The main circuit board of the system unit.
    - **Key Components on the Motherboard:**
        - Slot for processor chip
        - Slots for memory modules
        - Slots for adapter cards

2. The Processor (CPU)

- **Definition:** The processor, also called the Central Processing Unit (CPU), interprets and carries out the basic instructions that operate a computer.
- **Processor Variations:**
    - Dual-core processor
    - Multi-core processor
    - Quad-core processor
    - Octa-core processor
    - Deca-core processor (e.g., Intel i9)
- **Main Sub-Components of the Processor:**

    - **Control Unit (CU):** Directs and coordinates most of the operations in the computer.
    - **Arithmetic Logic Unit (ALU):** Performs arithmetic, comparison, and other logical operations.
- **System Data Flow:**
    - **Input Devices** send data into **Memory**.
    - **Memory** exchanges instructions, data, and information with the **Processor** (Control Unit and ALU).
    - **Memory** exchanges instructions, data, and information with **Storage Devices**.
    - **Memory** transmits finished information to **Output Devices**.

The Machine Cycle

For every instruction, a processor repeats a set of four basic operations that comprise a machine cycle:

1. **Step 1 (Fetch):** The control unit fetches the instruction and data from memory.
2. **Step 2 (Decode):** The control unit decodes the instructions and directs data to the ALU.
3. **Step 3 (Execute):** The ALU performs the required calculations or logical operations on the data.
4. **Step 4 (Store):** The results of the calculations are stored back into memory (subsequently rendered on output displays like a monitor).

3. Data Representation

- **Analog Signals:** Continuous signals that vary in strength and quality (e.g., soundwaves, analog thermometers, vinyl records).
- **Digital Signals:** Discrete signals that represent information using binary values (0s and 1s).
    - **Binary Digit 1:** Indicates presence of electronic charge (Electronic State: ON).
    - **Binary Digit 0:** Indicates absence of electronic charge (Electronic State: OFF).
    - Examples: CDs, MP3s, digital photos.

4. Memory

- **Definition:** Electronic components that store instructions waiting to be executed by the processor, data needed by those instructions, and the results of processed data.
- **Three Categories of Items Stored in Memory:**
    a. Application programs
    b. Operating system and other system software
    c. Data being processed and resulting information

Memory Capacities and Units

- **Kilobyte (KB or K):** ~1 thousand bytes (Exact: 1,024 bytes) — ~1/2 page of text.
- **Megabyte (MB):** ~1 million bytes (Exact: 1,048,576 bytes) — ~500 pages of text.
- **Gigabyte (GB):** ~1 billion bytes (Exact: 1,073,741,824 bytes) — ~500,000 pages of text.
- **Terabyte (TB):** ~1 trillion bytes (Exact: 1,099,511,627,776 bytes) — ~500,000,000 pages of text.
- **Petabyte (PB):** 1,024 Terabytes or 1,048,576 Gigabytes.
- **Exabyte (EB):** 1,024 Petabytes.
- **Zettabyte (ZB):** 1,024 Exabytes.
- **Yottabyte (YB):** 1,024 Zettabytes (1,208,925,819,614,629,174,706,176 bytes).

Memory Types: Volatile vs. Nonvolatile

- **Volatile Memory:** Loses its contents when computer power is turned off (e.g., RAM).
- **Nonvolatile Memory:** Retains its contents even when power is turned off (e.g., ROM, Flash memory/drives).

Random Access Memory (RAM)

- Installed on memory modules (such as dual inline memory modules - DIMMs) fitted into memory slots on the motherboard.
- **RAM Types:**
    - **Dynamic RAM (DRAM):** Standard dynamic memory chips.
    - **Static RAM (SRAM):** Faster, cache-oriented static memory.
    - **Magnetoresistive RAM (MRAM):** Uses magnetic charges instead of electrical charges.
- **DRAM Variations:**
    - **SDRAM (Synchronous DRAM):** Synchronized to the system clock; much faster than basic DRAM.

    - **DDR SDRAM:** Transfers data twice per clock cycle; faster than SDRAM.
    - **DDR2:** Second generation DDR; faster than DDR.
    - **DDR3:** Third generation DDR; designed for multi-core processors; faster than DDR2.
    - **RDRAM (Rambus DRAM):** Uses pipelining techniques; much faster than SDRAM.
- **RAM Allocation Recommendations:**
    - **2 GB or less:** Basic home/business use, word processing, web communication, personal finances.
    - **2 GB to 8 GB:** Advanced multimedia, accounting spreadsheets, voice recognition, video/audio editing, gaming, video conferences.
    - **8 GB and up:** Power users, CAD design, 3D modeling, professional graphics, web development.

ROM and Flash Memory

- **Read-Only Memory (ROM):** Memory chips storing permanent, non-modifiable instructions and data.
- **Programmable ROM (PROM):** Blank ROM chip that can be written to once; once programmed, contents are permanent.
- **Flash Memory:** Nonvolatile memory that can be erased and rewritten electronically.

5. Device Care and Preventive Maintenance

- Clean computer or mobile devices 1 to 2 times a year.
- Power down and unplug devices before beginning maintenance.
- Use compressed air to safely remove internal and keyboard dust.
- Clean cases with antistatic wipes; clean display screens using specialized screen solution and soft microfiber cloths.

6. Input Concepts and Peripherals

- **Definition:** Input is any data or instructions entered into the memory of a computer.
- **Forms of Input Instructions:**
    - **Programs:** A series of related instructions that direct computer operations.
    - **Commands:** Specific user directives that programs respond to.
    - **User Responses:** Answers provided by users to questions or prompts displayed by programs.
- **Input Devices:** Hardware components facilitating data entry:
    - Keyboards, mice, and touch-sensitive pads

    - Touch screens and graphic tablets
    - Stylus pens
    - Gamepads, steering wheels, and game controllers
    - Microphones, webcams, and digital cameras
    - Flatbed document scanners
    - Barcode readers and Optical Mark Recognition (OMR) scanners
    - RFID readers and Magnetic Stripe Card Readers
    - Magnetic Ink Character Recognition (MICR) readers
    - Biometric authenticators (fingerprint and hand scanners)

PART 2: DIGITAL SECURITY, ETHICS, AND PRIVACY

1. Foundations of Digital Security

- **Security:** Protection from hostile forces intending to compromise the normal state of being of a system; the ultimate goal of being safe from malicious actors.
- **Risk:** Any event or action that could cause loss of or damage to hardware, software, data, information, or processing capabilities. Risk must be assessed and managed, as 100% immunity is impossible.
- **The CIA Triad:**
    - **Confidentiality:** Safeguarding data from unauthorized access or disclosure.
    - **Integrity:** Ensuring information remains accurate, valid, and unaltered by unauthorized parties.
    - **Availability:** Ensuring legitimate users have timely, reliable access to critical data and services.

2. Cybercrimes & Categories of Threat Actors

- **Cybercrime:** An international breach of security involving deliberate, illegal actions.
- **Categories of Threat Actors:**
    - **Threat Actor:** The overarching umbrella term for anyone who maliciously interacts with a computer system.
    - **Script Kiddie:** An attacker lacking deep technical knowledge who utilizes pre-built hacking utilities created by others.
    - **Hacker:** A threat actor who circumvents computer security without authorization.
    - **Cracker:** An illegal intruder with explicit malicious intent to destroy data, steal assets, or sabotage networks.

    - **Hacktivist:** An individual breaching systems to promote socio-political movements or ideological causes.
    - **Cyberterrorist:** Attackers targeting national infrastructure (power grids, tech backbones) to compromise security and induce panic.
    - **Cyberextortionist:** Attackers holding systems or data hostage, demanding financial payouts to prevent data leakage or destruction.

3. Cyber Law & Digital Forensics

- **Republic Act 10175 (Cybercrime Prevention Act of 2012):** Legislation defining cybercrime offenses, laying down investigative rules, establishing preventative frameworks, and imposing legal penalties.
- **Digital Forensics:** The methodical discovery, extraction, collection, and forensic analysis of digital evidence on computers, mobile hardware, servers, and networks.

4. Internet and Network Attack Methodologies

- **Malware:** Contraction of "malicious software"; programs intended to disrupt system security.
    - *Case Study:* **WannaCry (2017)** — A widespread -worm hybrid that crippled hundreds of thousands of computers across 150+ countries, causing billions in damage.
- **Botnets:** A coordinated network of compromised machines ("zombies") controlled by an attacker.
    - *Applications:* DDoS floods, illicit cryptocurrency mining, automated troll networks.
    - *Case Study:* **Mirai Botnet (2016)** — Enslaved default-password IoT hardware to launch historic DDoS outages.
- **Denial of Service (DoS / DDoS):** Flooding a server or network infrastructure with excessive traffic to overwhelm resources and deny service to legitimate users.
    - *Case Study:* **Dyn Attack (2016)** — DNS infrastructure DDoS that knocked out Twitter, Netflix, Spotify, Reddit, PayPal, and major outlets across North America and Europe.
- **Backdoors:** Hidden bypass routines allowing attackers persistent access without triggering normal authentication.
    - *Case Study:* **SolarWinds Supply Chain Attack (2020)** — Malicious code planted in upstream updates gave intruders access to thousands of enterprise and government clients.

- **Spoofing:** Falsifying network headers, IP addresses, or domain interfaces to mask identity and appear authentic to victims.
- **Social Engineering:** Psychological manipulation designed to exploit human trust and induce individuals into revealing sensitive credentials or downloading malicious payloads (e.g., Phishing).

5. Defensive Countermeasures & Security Safeguards

- **Antivirus Utilities:** Active scanning, quarantine, and removal of malicious code.
- **Firewalls:** Filtering and monitoring inbound/outbound network traffic via strict rule-sets.
- **Virtual Private Networks (VPNs):** Encrypted tunneling protecting communication from eavesdropping.
- **Access Control:** Enforcing least-privilege principles and authenticating authorized personnel.
- **Redundant Data Backups:** Maintaining off-site or isolated file archives to survive and failures.
- **Rogue Network Awareness:** Avoiding "Evil Twin" hotspots designed to intercept sensitive wireless packets.
- **Digital Detox:** Reducing digital exposure and attack surface.

6. Information Privacy & Authentication

- **Strong Passwords:** Long, complex character sequences resistant to dictionary and brute-force attacks.
- **Two-Factor Authentication (2FA):** Multi-tiered identity verification requiring secondary approval factors.
- **Biometric Systems:** Hardware-based physical identification (fingerprints, facial geometry).
- **CAPTCHA:** Automated challenges to distinguish human interaction from automated bot traffic.
- **Encryption:** Cryptographic translation of readable plaintext into unreadable ciphertext across storage and transmission.

PART 3: INPUT & OUTPUT — ENTERING DATA AND PRODUCING INFORMATION (LESSON 7)

1. Understanding Input

- **Definition:** Any data or command entered into a computer or device, which the system then processes.
- **Forms of Input:**
  - Text
  - Pointing / Clicking
  - Touch
  - Voice
  - Scanning

2. Input Types & Modalities

- **Manual Input:**
  - Involves direct human physical interaction with an input device (e.g., typing, clicking, speaking).
  - Contrasted with automatic input generated by sensors or background signals.
  - *Manual Input Devices:* Keyboard, mouse, microphone, webcam.
- **Touch Input:**
  - Interacting directly with a touchscreen or touch-sensitive surface.
  - Detects contact from fingers, styluses, or specialized digital pens.
  - *Everyday Applications:* Self-service ordering kiosks, smartphones/tablets, GPS/navigation consoles.
- **Pen Input:**
  - Uses a stylus or digital pen on a flat surface or directly on a screen to write, sketch, or capture and digitize handwriting.
- **Digital Input:**
  - Devices enabling the capture and processing of video, audio, or digital data (e.g., voice input, video input, digital video technology).
- **Motion Input:**
  - A device containing a camera or motion sensors that detects body gestures and converts them into digital control signals.
  - *Entertainment:* Motion-controlled gaming (e.g., Nintendo Wii Remote).

    - *Military / Security:* AN/GAR-2 Military LKMD (Light Motion Detector Kit), providing early warning, situational awareness, and perimeter detection during combat missions.

3. Understanding Output

- **Definition:** Output is the result of processed information produced after a device interprets and computes input data (the actionable feedback provided to users).
- **Basic Types of Output:**
    - Text
    - Graphics
    - Audio
    - Video
- **Output Classification:**
    - **Physical Output (Hard Copy):** Tangible, permanent output that can be physically handled and kept (e.g., printed documents).
    - **Digital Output (Soft Copy):** Intangible electronic output displayed on screens or broadcast through audio speakers.

4. Assistive / Adaptive Technology & Accessibility

- **Goal of Accessibility:** Enable all individuals, including persons with disabilities, to interact with technology independently for work, learning, access to services, and entertainment.
- **Obligation to Provide Access:** Educational institutions, employers, and organizations are ethically and legally responsible for providing accessible input/output accommodations.
- **Adaptive Input Devices:**
    - Customized ergonomic/oversized button switches and adapted mouse controls.
    - Eyegaze tracking systems (e.g., EyeGaze Edge) allowing users to type and communicate via eye gaze movement.
- **Adaptive Output Devices:**
    - Refreshable Braille displays and terminals.
    - Braille embossers / Braille printers (e.g., Index Braille).
    - Optical text-to-speech reading pens.

5. Workspace Customization & Ergonomics

- **Productivity Gains:** Personalizing desktop settings, pin-mapping daily applications (e.g., spreadsheets), and using keyboard shortcuts streamline everyday workflows.
- **Ergonomics & Strain Reduction:** Calibrating screen brightness, font scaling, display contrast, and mouse tracking reduces eye strain, wrist tension, and bad posture (vital for intensive users such as accounting students).
- **Accessibility Awareness:** Familiarity with built-in accessibility tools (screen magnifiers, on-screen keyboards) promotes an inclusive workplace culture.

6. Environmental Responsibility & E-Waste Management

Practical, healthy technology habits:

5. **Radiation Shields:** Reduce close-contact electromagnetic exposure where applicable.
6. **Safe Screen Distance:** Maintain proper viewing distances to protect vision.
7. **Power Down Wireless Hardware:** Turn off unused Bluetooth/Wi-Fi peripherals when inactive.
8. **Recycle or Donate Old Devices:** Extend hardware lifecycles or use certified e-waste recyclers to prevent toxic landfill pollution.
9. **The 20-20-20 Rule:** Every 20 minutes, look at an object at least 20 feet away for at least 20 seconds to prevent digital eye strain.

PART 4: OPERATING SYSTEMS — MANAGING, COORDINATING & MONITORING RESOURCES (LESSON 8)

1. What is an Operating System (OS)?

- **Definition:** A set of programs that coordinates and manages all the activities and communications among computer or mobile hardware components.

2. Core Operating System Functions

- Starting and shutting down the computer or mobile device.
- Providing a user interface (UI).
- Managing programs and multitasking environments.

- Memory management and virtual memory swapping.
- Coordinating execution order of tasks and process spooling.
- Device configuration and driver communication.
- Monitoring real-time system performance.
- Establishing and managing internet and network connections.
- Providing file and storage media management.
- Managing automatic system updates.
- Administering network access and enforcing security/permissions.

3. Startup & Power State Management

- **Powering On:** Booting devices via physical power switches across desktops, laptops, and mobile hardware.
- **Sleep Mode:** Saves open documents and running applications to RAM, powers down unnecessary hardware, and keeps the computer in an ultra-low-power standby state.
- **Hibernate Mode:** Saves all open documents and active memory states onto the internal secondary storage (hard drive/SSD) and cuts power completely.

4. User Interfaces (UI)

- **Graphical User Interface (GUI):** Allows users to interact intuitively through visual menus, icons, buttons, tiles, and graphical windows (e.g., Windows Desktop, macOS Launchpad/Dock).
- **Command-Line Interface (CLI):** Requires users to type exact keywords, abbreviations, and parameters following strict spelling and syntax rules at a command prompt (e.g., Unix/Linux Bash shell).

5. Program and Multitasking Management

- **Single-tasking vs. Multitasking:** Running a single application at a time versus handling multiple programs concurrently.
- **Foreground vs. Background Processing:**
    - *Foreground Applications:* The active program currently receiving user focus and input.
    - *Background Applications:* Programs running invisibly behind the scenes (e.g., music streaming, antivirus scanners, print queues).
- **Single-user vs. Multiuser:** Supporting a solitary operator versus arbitrating simultaneous access for hundreds of network users.

6. Memory Management & Virtual Memory

- **Optimization:** Allocates dynamic RAM sectors between the OS kernel and running apps.
- **Virtual Memory:** A portion of secondary storage configured to act as additional temporary RAM.
    - **Step 1 (Page Out):** The OS transfers least-recently-used data blocks (pages) from RAM into a swap file on the disk drive.
    - **Step 2 (Page In):** When those swapped-out instructions are requested again, the OS retrieves them from the swap file and places them back into physical RAM.

7. Task Coordination & Print Spooling

- The OS regulates scheduling priorities for incoming tasks.
- **Print Spooling:** Print jobs are routed to temporary storage buffers (on disk or memory) forming a print queue, freeing the source program immediately while the printer outputs pages at its mechanical speed.

8. Device Configuration & Drivers

- **Driver:** A specialized low-level software program that instructs the OS on how to communicate with a specific peripheral.
- **Plug and Play (PnP):** Automatic detection, driver assignment, and configuration of peripheral devices upon connection.

9. Performance Monitoring & Network Integration

- **Performance Monitor:** Diagnostic utilities (e.g., Task Manager, Activity Monitor) displaying real-time metrics on CPU utilization, memory pressure, disk I/O, network throughput, and running processes.
- **Network Connectivity:** Built-in connection wizards enabling seamless configuration of Wi-Fi hotspots, broadband connections, enterprise VPNs, and workplace domains.

10. File, Disk, and System Management Utilities

- **File Manager:** Displays files, directories, folder hierarchies, and supports copy, move, rename, delete, and sort actions.
- **Search:** Locates files and applications using user-defined criteria or filenames.
- **Image Viewer:** Displays, basic-edits, prints, and previews image formats.

- **Uninstaller:** Completely removes applications and scrubs leftover registry/system files.
- **Disk Cleanup:** Scans for and safely purges temporary cache files, download archives, and system bloat.
- **Disk Defragmenter:** Realigns fragmented file sectors on magnetic hard disks into contiguous spaces for faster access.
- **Screen Saver:** Plays animations or blanks the screen after user inactivity.
- **File Compression:** Encodes data to shrink overall file size for storage or transmission.
- **PC Maintenance:** Diagnoses system crashes, corrects disk directory errors, and optimizes responsiveness.
- **Backup and Restore:** Duplicates local volumes to remote/secondary media to prevent data loss.

11. Categories of Operating Systems

- **Desktop Operating Systems:** Comprehensive systems engineered for desktops, laptops, and tablets:
    - *Windows:* Widely used commercial OS featuring start menus, Live Tiles, and broad software compatibility.
    - *macOS:* Apple's proprietary UNIX-based desktop OS with Launchpad, Dock, and tight hardware integration.
    - *UNIX:* Multitasking, multi-user operating system developed in the early 1970s.
    - *Linux:* Open-source, UNIX-like multitasking operating system maintained globally.
    - *Chrome OS:* Lightweight Linux-based system designed to work primarily through cloud and web applications.
- **Server Operating Systems:** Multiuser operating systems designed to manage networks, high client concurrency, and shared resources:
    - *Windows Server:* Enterprise server environment offering on-premise and hybrid cloud administration.
    - *OS X Server:* Server software suite designed for Apple hardware networks.
    - *UNIX / Linux:* Highly stable, secure operating systems powering the majority of Internet web servers, cloud clusters, and supercomputers.
- **Mobile Operating Systems:** Embedded operating systems residing in firmware on portable devices:
    - *Android:* Google’s open-source, Linux-based mobile OS for phones and tablets.

    - *iOS:* Apple's proprietary closed-source mobile OS tailored specifically for iPhones and iPads.
    - *Windows Phone:* Microsoft's proprietary mobile OS featuring a Live Tile interface.

12. Advanced Topics: Virtualization, Licensing & Security

- **Virtual Machines (VMs):** Software-based emulated computers running on physical host machines. A host system can run isolated guest operating systems concurrently (e.g., running Linux on Windows using VMware or VirtualBox).
- **OS Licensing:** The governing legal contract (EULA/open-source licenses) defining distribution, usage rights, and modification terms. Valid licensing ensures regulatory compliance and software updates.
- **Securing the OS:**
    - Install only trusted, necessary software packages.
    - Enforce strict user privilege policies and role-based access control.
    - Regularly apply critical security patches and OS kernel updates.
    - Run persistent performance and security monitoring tools.
    - Maintain routine, tested backups.